

Problem

A balanced delta-connected load has a phase current

$$\mathbf{I}_{AC} = 5 \angle -30^\circ \text{ A}.$$

(a) Determine the three line currents assuming that the circuit operates in the positive phase sequence.

(b) Calculate the load impedance if the line voltage is $\mathbf{V}_{AB} = 440 \angle 0^\circ \text{ V}$.

Step-by-step solution

Step 1 of 6

(a)

Consider the value of the phase current $\mathbf{I}_{AC}$.

$$\mathbf{I}_{AC} = 5 \angle -30^\circ \text{ A}$$

Compare with $\mathbf{I}_p \angle \phi^\circ$.

The values of $\mathbf{I}_p$ and ϕ° are 5 A and -30° respectively.

Step 2 of 6

Calculate the value of the line current $\mathbf{I}_{cC}$.

$$\mathbf{I}_{cC} = \sqrt{3} \mathbf{I}_p \angle \phi - 30^\circ$$

Substitute -30° for ϕ° and 5 A for $\mathbf{I}_p$.

$$\begin{aligned} \mathbf{I}_{cC} &= \sqrt{3}(5) \angle -30^\circ - 30^\circ \\ &= 8.66 \angle -60^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current $\mathbf{I}_{cC}$ is $\boxed{8.66 \angle -60^\circ \text{ A}}$.

Compare with $\mathbf{I}_L \angle \theta^\circ$.

The values of $\mathbf{I}_L$ and θ° are 8.66 A and -60° respectively.

Step 3 of 6

In positive sequence, the line current $\mathbf{I}_{aA}$ leads $\mathbf{I}_{bB}$ which in turn leads $\mathbf{I}_{cC}$.

Calculate the value of the line current $\mathbf{I}_{aA}$.

$$\mathbf{I}_{aA} = \mathbf{I}_L \angle \theta - 120^\circ$$

Substitute -60° for θ° and 8.66 A for $\mathbf{I}_L$.

$$\begin{aligned} \mathbf{I}_{aA} &= 8.66 \angle -60^\circ - 120^\circ \\ &= 8.66 \angle -180^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current $\mathbf{I}_{aA}$ is $\boxed{8.66 \angle -180^\circ \text{ A}}$.

Step 4 of 6

Calculate the value of the line current I_{bB} .

$$I_{bB} = I_L \angle \theta + 120^\circ$$

Substitute -60° for θ° and 8.66 A for I_L .

$$\begin{aligned} I_{bB} &= 8.66 \angle -60^\circ + 120^\circ \\ &= 8.66 \angle 60^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_{bB} is $\boxed{8.66 \angle 60^\circ \text{ A}}$.

Step 5 of 6

(b)

Calculate the value of the current I_{AB} .

$$I_{AB} = I_p \angle \phi - 120^\circ$$

Substitute -30° for ϕ° and 5 A for I_p .

$$\begin{aligned} I_{AB} &= 5 \angle -30 - 120^\circ \\ &= 5 \angle -150^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_{AB} is, $5 \angle -150^\circ \text{ A}$.

Step 6 of 6

Calculate the value of the load impedance Z_Δ .

$$Z_\Delta = \frac{V_{AB}}{I_{AB}}$$

Substitute $5 \angle -150^\circ \text{ A}$ for I_{AB} and $440 \angle 0^\circ \text{ V}$ for V_{AB} .

$$\begin{aligned} Z_\Delta &= \frac{440 \angle 0^\circ}{5 \angle -150^\circ} \\ &= 88 \angle 150^\circ \Omega \end{aligned}$$

Therefore, the value of the load impedance Z_Δ is, $\boxed{88 \angle 150^\circ \Omega}$.